



English photographer Eadweard Muybridge's pictures showed the exact position of a trotting horse's legs for the first time. The original pictures shot in 1877 did not survive. The ones seen here are from 1878.

## ALEXANDER GRAHAM BELL (1847–1922)



Alexander Graham Bell was born in 1847 in Edinburgh, Scotland. In 1870, he first moved to Ontario, Canada, and then Boston, US, where he began his career as an inventor. Both Bell's mother and wife suffered hearing impairments, inspiring his interest in speech and hearing. This interest led him to invent the microphone and the telephone.

fiber bundle called the arcuate fasciculus. Damage to these fibers can leave people able to understand language, but with speech that makes no sense.

German physiologist **Wilhelm Kühne** (1837–1900) discovered and named the pancreatic enzyme **trypsin** in 1876. He also coined the term **enzyme**, which

refers to large biological molecules that act as catalysts and control the rate of chemical reactions in the cells of living things (see 1893–94).

In 1877, German bacteriologist **Robert Koch** (1843–1910) grew *Bacillus anthracis*—the organism that causes the infectious disease **anthrax**—in the laboratory and injected it into animals to induce the disease. This was the first bacterium shown to be the cause of a disease.

In the world of psychology, **Charles Darwin** published *Biographical Sketch of an Infant*, a 10-page book based on his observations of his newborn son, from 37 years earlier. Darwin speculated that his son went through the same stages of

learning that his ancestors had for centuries, and that “each individual somehow retains rudiments of the long evolutionary past.”

In 1877, English photographer **Eadweard Muybridge** (1830–1904) made a **camera** with a **shutter speed of only one-thousandth of a second**. With this and an ultrasensitive photographic plate he was able to “freeze” the motion of a trotting horse in a picture. The picture caused a sensation, proving that all four of the horse's legs left the ground at once.

# 274 THE NUMBER OF AMINO ACIDS IN TRYPSIN

American astronomer **Asaph Hall** (1829–1907) discovered the two moons of Mars in 1877. He spotted **Deimos**, the smaller moon, on August 12, and **Phobos** on August 18. Hall also worked out the mass of Mars, the orbits of moons, and measured the rotation of Saturn.

Italian astronomer **Giovanni Schiaparelli** (1835–1910) was also studying Mars in 1877. He reported seeing **canali** on the planet. This word, simply meaning “channels,” was mistranslated to “canals,” giving rise to a frenzy of speculation that Mars was inhabited by intelligent beings. It was later discovered that these markings were optical illusions.

**Bell's electric telephone**  
Used as both transmitter and receiver, this early telephone converted sound vibrations into electrical signals (and vice versa).



**May 1876** Nikolaus Otto builds the first four-stroke piston engine

**1876** Karl Wernicke locates **aphasia** at a different part of the brain from Broca

**1876** Wilhelm Kühne discovers the pancreatic enzyme **trypsin**

**1876** Robert Koch identifies the cause of **anthrax**

**1877** Charles Darwin publishes the first book on **child psychology**

**1877** Swiss zoologist Hermann Fol observes **spermatozoan** of starfish penetrating the egg

**August 1877** American astronomer Asaph Hall discovers the **two moons** of Mars

**1877** Austrian physicist Ernst Mach publishes a paper on **supersonic motion**

**1877** Giovanni Schiaparelli observes what he believes are **channels** on Mars

**1877** American inventor Thomas Edison invents the **phonograph**